Scratch Primary Supplemental Lessons Tutors Manual

Diving Deeper With Scratch

Table of Contents

Title	Page
The Aim of This Lesson Series	4
How to Use This Manual	5
Lesson Format	6
Lesson 1 Skills – Getting Started	7
Lesson 1 Plan – Getting Started	8
Lesson 2 Skills – The Cat and Bat	9
Lesson 2 Plan – The Cat and Bat	10
Lesson 3 Skills – Help the Cat Escape	11
Lesson 3 Plan – Help the Cat Escape	12
Lesson 4 Skills – Motion and Direction	13
Lesson 4 Plan – Motion and Direction	14
Lesson 5 Skills – The XY Coordinate System	15
Lesson 5 Plan – The XY Coordinate System	16

Table of Contents continued

Title	Page
Lesson 6 Skills – Polygons and Flowers	17
Lesson 6 Plan – Polygons and Flowers	18
Lesson 7 Skills – Good Guy / Bad Guy Game 1	19
Lesson 7 Plan– Good Guy / Bad Guy Game 1	20
Lesson 8 Skills – Good Guy / Bad Guy Game 2	21
Lesson 8 Plan – Good Guy / Bad Guy Game 2	22

The Aim of This Lesson Series

These lessons provide a more in-depth and detailed description of concepts introduced in lessons 1-10 found in the Primary Resources section of the scratch.ie website:

http://scratch.ie/primary/lessonplans

1. Under the Sea
2. In the Jungle
3. Battle of the Bands
4. A Cartoon about Me

etc…

How to Use This Manual

This manual contains the following outlines for each lesson in the series:

- Lesson Skills
  - A summary of the expected learning outcomes for the given lesson consisting of:
    - Computer science and mathematical concepts.
    - Specific Scratch skills.
- Lesson Plan
  - An overview of the sequence and timing of each lesson. This is broken down into 5 – 15 minute intervals. This serves to keep each lesson to within the target time of 50 minutes.
- Project Resources
  - Links to external projects that are used in the lesson.

Lesson Format

- This lesson series consists of eight “ready made” lessons, each of which has been planned to take just under 1 hour.
- Each lesson demonstrates key concepts using concrete examples in a step by step format. The prescriptive format is appropriate because:
  - The structured lessons provide a compliment to the explorative and experimental activities that the students will be doing on their own.
  - Volunteers who are delivering the lessons may not have extensive teaching backgrounds, therefore, the lessons provide structure and a known outcome.
  - IT professionals who will be volunteering one hour out of their work day to visit a school may not have the time to prepare lessons. The ready made lessons offer a time savings.
  - These lessons provide a common reference point for the entire class when applying the concepts to their own projects during the school year.

Lesson 1 Skills: Getting Started

- In lesson 1 we use the context of creating an animation to introduce the following concepts:
  - Sequential execution of instructions
  - Loops
  - Multi-threading
- Scratch skills:
  - Dragging blocks to make scripts
  - Animations by changing costumes
  - Adding/ changing a background
- Project Resources
  - The completed project is at: http://scratch.mit.edu/projects/12624412/

Lesson 1 Plan: Getting Started

Activity	Time	Pages
Introduce the Scratch application • Navigate to Scratch website or start up "offline editor" • Sign in to Scratch. Save project as individual name.	5 min	2-4
Introduce the computer as a machine waiting for instructions • Introduce the speech block as the first example of an instruction • Create more speech blocks to demonstrate sequential execution	10 min	5-12
Introduce the forever loop • Introduce animation by changing costumes • Introduce the wait statement. Guide class towards discovering loop symmetry. • Introduce movement using the move block.	15 min	13-20
Control the speed of the animation using numbers	5 min	21
Introduce Multi-Threading – Use green flag to start all	5 min	22
Fun Stuff – if time allows • Introduce "If on edge, bounce" block in the forever loop. • Change the direction of the sprite and see what happens. • Add a background.	5 min	23-25
Save project for use in next lesson	2 min	26

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Lesson 2 Skills: The Cat and Bat

- In lesson 2 we create a second character to our story to introduce the following concepts:
  - Communication between sprites
  - Story creation
  - Event timing
- Scratch skills:
  - Key press events
  - Broadcasting
  - Hide and show sprites

- Project Resources
- The lesson can start with this project: http://scratch.mit.edu/projects/12624412/
- The completed project is at: http://scratch.mit.edu/projects/12624773/
  - Demonstrate the completed project at the beginning of class.

Lesson 2 Plan: The Cat and Bat

Activity	Time	Pages
Start with the animation created in lesson 1 • If students don't have theirs, they can use this one: http://scratch.mit.edu/projects/12624412/ • Discuss the story plan. • Demo completed project: http://scratch.mit.edu/projects/12624773/	5 min	2
Add a second sprite – the bat. Distribute a copy of the finished "Hello World" project to those who need it: http://scratch.mit.edu/projects/12624412/ • Make the bat appear suddenly and say, "Boo"!	10 min	3,4
Make the bat flap it wings	5 min	5
Make the cat run away from the bat without broadcasting • Test the bat and cat actions individually. • Put the story together. • Show limitation of wait delay by making the bat appear with the space key	10 min	6-10
Introduce Broadcasting • Make the bat scare the cat into running away using broadcasting	10 min	11-14
Run the Story. • Save your work	5 min	15-16

Lesson 3 Skills: Help the Cat Escape

- In lesson 3 we continue the story by having the cat escape to another scene. This covers the following concepts:
  - Communication between objects (sprites and backgrounds)
  - If statement – conditional logic
  - Sensing - Edge detection
  - Polling
  - Boolean logic – True / False
  - Multi – threading
  - Unit testing and debugging
  - Events
- Scratch skills:
  - Broadcasting to multiple objects
  - Sensing
- Project Resources
  - **The lesson must start with this project:** http://scratch.mit.edu/projects/12624773/
  - The completed project is at http://scratch.mit.edu/projects/12625222/
    - Demonstrate the completed project at the beginning of class

Lesson 3 Plan: Help the Cat Escape

Activity	Time	Pages
Start up • Open previous projects and discuss the story idea. • **Students must use their previous project as the lesson starting point!!** • If students don't have their previous project they can use this one: http://scratch.mit.edu/projects/12624773/	5 min	2-5
Introduce sensing : the "touching <_>?" statement • Talk about possible values i.e. True, False, other. • Test the touching script.	15 min	6-9
Introduce the "If" statement • Discuss binary logic and decision making.	10 min	10-15
Introduce polling with the "forever" loop • Debug script and discover how polling works.	10 min	16-20
Change the scene by using broadcasting • Escape from the bat.	15 min	21-26
Save your work!	2 min	27
Fun Stuff – if time allows • Make the cat enter a different scene every time he touches the edge.	5 min	25

Lesson 4 Skills: Motion and Direction

- In lesson 4 we use the move block and direction to introduce the following concepts:
  - Moving in positive and negative steps
  - Controlling movement with arrow keys – user input
  - Direction and angles
- Scratch skills:
  - The “Move” block
  - Direction blocks – setting direction
  - Key press blocks
  - Setting rotation style
  - How angle and direction work together.
  - Copying scripts between sprites.
- Project Resources
  - The completed project is at http://scratch.mit.edu/projects/12624196/
    - Demonstrate the completed project at the beginning of class

Lesson 4 Plan: Motion and Direction

Activity	Time	Pages
Introduce the "move" block: • Start a new project. • Experiment with clicking on the "move" block. • Hook up the "move" block to the right arrow key.	5 min	2-4
Negative Numbers • Hook up the "move" block to the left arrow key. • Figure out how to move left with the left arrow key. • Discuss negative numbers and how they relate to left movement. • Show the number line and it's relationship to L and R movement.	10 min	5-8
Direction • Move up and down. Discuss how this can be done. • Experiment with choosing different directions. • Introduce direction and angles. Show the circle and all it's degrees. • Make the sprite face the direction it is moving	10 min	9-14
Rotation • Introduce the rotation styles.	10 min	15, 16
Animation • Make the character walk (or fly or swim) when moving. • Choose other characters which have animation costumes.	15 min	17, 19

Lesson 5 Skills: The XY Coordinate System

- In lesson 5 we explore the XY coordinate system and introduce the following concepts:
  - The stage size
  - The pixel
  - Movement along the x axis
  - Movement along the y axis
- Scratch skills:
  - XY position in 2 dimensional coordinate space
  - The “go to x_ y_” block
  - The “glide to x_ y_” block
  - Combining x and y to move in four directions up, down, right, left
- Project Resources
  - Demonstrate or have the class play with the “xyPosition” project: http://scratch.mit.edu/projects/12776565/
  - Add scripts to enhance the “Ant Farm” game: http://scratch.mit.edu/projects/12778304/

Lesson 5 Plan: The XY Coordinate System

Activity	Time	Pages
Scratch Stage size = 480 x 360 pixels • Describe the pixel • Ask students where do they think 0 is placed.	5 min	2,3
The X number line : • Show positive, negative, 0. Right and left maps to + and – • Hook up L and R arrow keys to corresponding changes in X	10 min	4 - 7
The Y number line • Show positive, negative, 0. Up and down maps to + and –	5 min	8
The XY coordinate system • Hook up all 4 arrow keys and move in the corresponding direction by changing x and y. • Play with the pen, make a drawing program. • Move the a sprite around the stage using go to x: y: and the glide blocks	15 min	9 - 15
Play with "xyPosition" demo. See how x and y work together http://scratch.mit.edu/projects/12776565/	5 min	16
Take the "Ant Farm" project and make the ant go through the maze http://scratch.mit.edu/projects/12778304/ • Move by using the arrow keys • Move by creating a script that makes the ant move automatically.	10 min	17-20

Lesson 6 Skills: Polygons and Flowers

- In lesson 6 we learn about polygons, how to draw flowers and introduce the following concepts:
  - The circle described in degrees
  - Repeat loops and the number of iterations to make in a complete circle.
  - Polygons and the number of degrees in the angle of their vertices
  - Nested repeat loops
- Scratch skills:
  - Repeat loops
  - The pen
  - Drawing on colored backgrounds
- Project Resources
  - Demonstrate the “Flowers” project: http://scratch.mit.edu/studios/254501/
  - Demonstrate the “Polygon Creator” project: http://scratch.mit.edu/projects/12990763/

Lesson 6 Plan: Polygons and Flowers

Activity	Time	Pages
Describe direction and the circle	5 min	2 - 4
Draw a square: • Draw one corner • Draw the remaining corners • Introduce the “repeat” loop	10 min	5 - 7
Polygons • Show how angles in vertices are related to the degrees in the circle divided by the number of sides	10 min	8 - 12
Draw polygons in Scratch • Experiment with the number of sides • Play with the pen, make multi-colored polygons. • Demo “Polygon Creator” project: http://scratch.mit.edu/projects/12990763/	10 min	13
Draw flowers in Scratch • Place the polygon loop inside of another repeat loop. • Draw several flowers inside each other • Draw flowers in various places all over the stage • Demo “Flowers” project: http://scratch.mit.edu/studios/254501/	10 min	14, 15

Lesson 7 Skills: Good Guy / Bad Guy Game 1

- In lesson 7 we create a game and introduce the following concepts:
  - The design process
  - Variables and Data
  - Incrementing and decrementing variables
  - Closing “Game Over” state – win or lose.
- Scratch skills:
  - Variables – how to create and use them.
  - Sprite creation
- Project Resources
  - The completed lesson 7 project: http://scratch.mit.edu/projects/13024872/

Lesson 7 Plan: Good Guy / Bad Guy Game 1

Activity	Time	Pages
Discuss the game design	5 min	2
Create the characters: • The good guy • The bad guy	10 min	3 - 7
Touch detection • Discuss score change on a “touching” event • Create the touch detection script • Test the touch detection script	10 min	8 - 12
Keeping score • Discuss the scoring objective • Create the score variable	10 min	13-17
Add the scoring script to the game	10 min	18,19
Play the game • Discuss ways to improve the game	5 min	20,21

Lesson 8 Skills: Good Guy / Bad Guy Game 2

- In lesson 8 we continue working on our game from the previous lesson and introduce the following concepts:
  - The design process
  - Logic
  - Incrementing and decrementing variables
  - Closing “Game Over” state – win or lose.
  - Random Numbers
- Scratch skills:
  - Variables – how to create and use them.
  - Sprite creation
  - Random placement of sprites on the stage.
- Project Resources
  - The completed lesson 7 project: http://scratch.mit.edu/projects/13024872/

Lesson 8 Plan: Good Guy / Bad Guy Game 2

Activity	Time	Pages
Discuss new features to add to the game • Continue with the game from the previous lesson. If not available use this one: http://scratch.mit.edu/projects/13024872/	5 min	2,3
Ending the game: • Create the scripts • Create the "You Lose" game over screen	20 min	4 - 12
Winning the game: • Create score change on a "touching" event • Create the touch detection script • Test the touch detection script • Create the "You Win" game over screen	20 min	13-19
Making the game more challenging • Use random numbers to change location of "health" sprite	5 min	20,21